

3.2–3.3 Nullspace, Special Solutions, and Complete Solutions

Concise exam revision notes for homogeneous and nonhomogeneous systems.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Row reduction separates a linear system into forced variables and free variables. Pivot variables are forced by the equations; free variables create the directions in the nullspace.

The two central systems are:

$$Ax = 0$$

and

$$Ax = b.$$

The first finds input directions that disappear under A . The second asks whether the right side b can be built from the columns of A .

2 Big Picture

Let A be an $m \times n$ matrix.

Question	Meaning	Output
$Ax = 0$	Which input vectors are sent to zero?	The nullspace $N(A)$.
$Ax = b$	Can the columns of A produce this b ?	Zero, one, or infinitely many solutions.
Row reduction	Reveals pivots and free variables.	The structure of all solutions.

The rank r is the number of pivots. The number of free variables is

$$n - r.$$

Exam trigger. If a problem says “find all solutions”, expect an answer of the form

$$x = x_p + x_n.$$

Here x_p is one particular solution and x_n runs through the nullspace.

3 Nullspace vs Column Space

Space	Definition	Lives in	What it tells you
Nullspace $N(A)$	All x such that $Ax = 0$.	$\mathbb{R}^n$	Directions that do not change Ax .
Column space $C(A)$	All linear combinations of the columns of A .	$\mathbb{R}^m$	Right sides b for which $Ax = b$ is solvable.

Why the dimensions differ:

- x has one entry per column of A , so $x \in \mathbb{R}^n$.
- Ax and b have one entry per row of A , so they live in $\mathbb{R}^m$.
- Therefore $N(A) \subseteq \mathbb{R}^n$, while $C(A) \subseteq \mathbb{R}^m$.

Common mistake. Do not place the nullspace in $\mathbb{R}^m$. The nullspace contains input vectors, so its vectors have length n .

4 Free Variables and Special Solutions

Free variables appear in columns without pivots. Each free variable gives one independent nullspace direction.

4.1 Solving pattern

1. Row-reduce A to echelon or reduced echelon form.
2. Mark pivot variables and free variables.

3. Set one free variable to 1 and all other free variables to 0.
4. Solve for the pivot variables.
5. The resulting vector is one special solution.
6. Repeat once for each free variable.

If the free variables are x_3 and x_4 , use

$$(x_3, x_4) = (1, 0) \quad \text{and} \quad (x_3, x_4) = (0, 1).$$

These choices produce two special solutions s_1 and s_2 .

Core idea. The special solutions form a basis for the nullspace:

$$N(A) = \text{span}\{s_1, \dots, s_{n-r}\}.$$

Exam trigger. If $n > m$, then $r \leq m < n$. There must be at least one free variable, so $Ax = 0$ has a nonzero solution.

5 Complete Solution

For $Ax = b$, first check consistency. If the system is consistent, all solutions are

$$x = x_p + x_n.$$

Part	Equation	Role	How to find it
Particular solution x_p	$Ax_p = b$	One fixed solution.	Often set free variables to 0.
Nullspace part x_n	$Ax_n = 0$	Directions added to x_p .	Combine special solutions.

If $s_1, \dots, s_k$ are special solutions, then

$$x = x_p + c_1 s_1 + c_2 s_2 + \dots + c_k s_k.$$

The constants $c_1, \dots, c_k$ are arbitrary.

Why this works:

$$A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b.$$

Adding a nullspace vector keeps the same right side.

Common mistake. Do not multiply the particular solution by an arbitrary constant. The free constants belong only to nullspace directions.

6 Rank, Pivots, and Free Variables

Quantity	Meaning	Exam use
$r = \text{rank}(A)$	Number of pivots.	Counts independent columns.
Pivot variables	Variables solved from equations.	Forced by the system.
Free variables	Variables chosen freely.	Create special solutions.
$n - r$	Number of free variables.	Dimension of $N(A)$.

The rank-nullity count used here is

$$\dim N(A) = n - r.$$

Exam memory. Full column rank controls uniqueness. Full row rank controls existence. Square full rank gives both.

For $Ax = 0$, row operations do not change the solution set, so it is enough to solve

$$Rx = 0.$$

For $Ax = b$, reduce the augmented matrix $[A \ b]$. A zero coefficient row with a nonzero augmented entry means no solution.

7 Full Column Rank vs Full Row Rank

Case	Condition	Nullspace	Behavior of $Ax = b$
Square full rank	$r = m = n$	Only 0.	Exactly one solution for every b .
Full column rank	$r = n \leq m$	Only 0.	Zero or one solution.
Full row rank	$r = m \leq n$	Dimension $n - m$.	At least one solution for every b .
Not full rank	$r < m$ and $r < n$	Dimension $n - r$.	Zero or infinitely many solutions.

7.1 Full column rank

Every column has a pivot.

- No free variables.
- $N(A) = \{0\}$.
- If a solution exists, it is unique.
- Some right sides b may still be outside the column space.

7.2 Full row rank

Every row has a pivot.

- The column space is all of $\mathbb{R}^m$.
- $Ax = b$ is solvable for every $b \in \mathbb{R}^m$.
- If $n > m$, free variables make the solution infinite.

Common mistake. Full column rank gives uniqueness, not automatic existence. Full row rank gives existence for every b , not uniqueness.

8 Rank-One Matrix Example

A rank-one matrix has the form

$$A = uv^T.$$

Then

$$Ax = u(v^T x).$$

If $u \neq 0$, then $Ax = 0$ is equivalent to

$$v^T x = 0.$$

So the nullspace consists of vectors perpendicular to v .

Example:

$$A = \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 10 \end{bmatrix}.$$

The nullspace condition is

$$x_1 + 3x_2 + 10x_3 = 0.$$

Free choice	Resulting special solution
$x_2 = 1, x_3 = 0$	$(-3, 1, 0)$
$x_2 = 0, x_3 = 1$	$(-10, 0, 1)$

Therefore

$$N(A) = \text{span}\{(-3, 1, 0), (-10, 0, 1)\}.$$

Common mistake. For $A = uv^T$, the nullspace condition is $v^T x = 0$, not $u^T x = 0$. The vector v matches the input dimension.

9 Common Mistakes

Mistake	Correct exam habit
Putting $N(A)$ in $\mathbb{R}^m$.	$N(A) \subseteq \mathbb{R}^n$, because x has n entries.
Counting rows to find special solutions.	Count free variables: $n - r$.
Giving only x_p for $Ax = b$.	Add the whole nullspace: $x = x_p + x_n$.
Scaling x_p by a free constant.	Only special solutions get arbitrary constants.
Treating full column rank as automatic solvability.	It gives uniqueness if a solution exists.
Treating full row rank as uniqueness.	It gives existence for every b ; free variables may remain.
Ignoring inconsistent rows.	Check $[A \ b]$ before writing a solution.

10 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Row-reduce and find the rank r .
3. Mark pivot variables and free variables.
4. For $Ax = 0$, create one special solution per free variable.
5. Check that the number of special solutions is $n - r$.
6. For $Ax = b$, reduce $[A \ b]$ and check consistency.
7. Find one particular solution x_p .
8. Add the nullspace:

$$x = x_p + c_1 s_1 + \cdots + c_k s_k.$$

9. Check dimensions: $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$.
10. State the outcome directly: no solution, one solution, or infinitely many.